

GitHub: <https://github.com/mr-uyghur/tutorconnect-cmpe172-milestone1>

Video: https://drive.google.com/file/d/1u3FaMFmUwonOc5w_u8asEpUoQraFAZew/view?usp=sharing

Milestone 1 - Proposal

Ali Halmamat | TutorConnect is a tutoring appointment system. Students find one-hour sessions with tutors for subjects such as Java and SQL. Tutors offer time slots and see their bookings.

Roles, entities, and technology

Customer: student. **Provider/admin:** tutor. The five main entities are users, providers, services, availability_slots, and appointments. The project uses Java, Spring Boot, and SQL through JDBC; it uses no ORM. It realizes classic enterprise standards on a modern stack: Spring Boot for J2EE-style applications and REST for the role of CORBA/distributed objects.

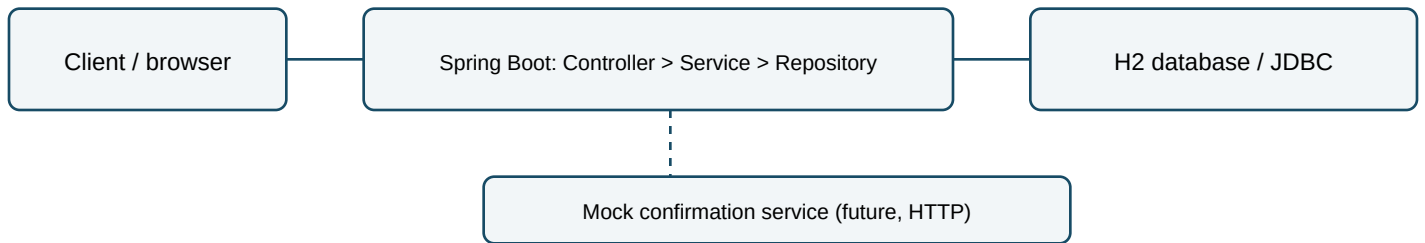
Twelve planned features

1. Student login.
2. Browse, filter by tutor/subject/date, and paginate available slots with SQL LIMIT/OFFSET.
3. Book a slot for its tutoring service.
4. View my upcoming and past appointments.
5. Cancel my own appointment.
6. Tutor login.
7. Tutor creates and removes service-linked availability slots.
8. Tutor views appointments booked with them.
9. Send booking and cancellation confirmations through a mock external service.
10. Add logging, a health endpoint, and one metric.
11. Add a conversational booking agent and an autonomous agent.
12. Add grounded question answering over a small knowledge base.

Milestone plan: Features 1-8 are for Milestone 2, 9-10 for Milestone 3, and 11-12 for Milestone 4. This first milestone implements the layered skeleton and two database reads only.

Architecture and relational schema

Block diagram and request flow



For **GET /slots**, Spring MVC routes the request to SchedulingController. It calls SchedulingService, which asks SlotRepository to run the SQL query. The query selects future slots with no BOOKED appointment. The repository fills SlotDto objects, which return through the service and controller as JSON. **GET /** follows the same path, using CatalogRepository to read tutors and services into HomeDto.

Page Controller vs Front Controller

A Page Controller organizes code around individual server-rendered pages, such as Thymeleaf pages. A Front Controller receives requests in one place and dispatches them. TutorConnect uses Spring MVC DispatcherServlet as the Front Controller for JSON requests; the four wireframes show planned client screens.

Relational schema (PK = primary key; FK = foreign key)

users: user_id PK; full_name; email UNIQUE; password_hash (nullable); role CUSTOMER/PROVIDER.

providers: provider_id PK; user_id FK → users.user_id and UNIQUE; bio.

services: service_id PK; name UNIQUE; description.

availability_slots: slot_id PK; provider_id FK → providers.provider_id; service_id FK → services.service_id; starts_at; ends_at. UNIQUE(provider_id, starts_at).

appointments: appointment_id PK; customer_id FK → users.user_id; slot_id FK → availability_slots.slot_id; status BOOKED/CANCELLED; created_at; generated active_slot_id UNIQUE.

Each slot begins on a whole hour and ends exactly one hour later. All fields are required except password_hash and the generated active_slot_id. The full SQL is in *schema.sql*; the ER diagram is a separate PDF.

Database rules and Milestone 1 code

Relationships and keys

A user can have zero or one provider profile. Each provider and each service can have many slots; each slot has one provider and one service. A student can have many appointments, and each appointment belongs to one student and one slot. A slot can have several past appointment records, but only one active booking. Every table has its own primary key, so there are no classical weak entities.

How double-booking is prevented

For a BOOKED appointment, the generated **active_slot_id** equals its slot_id. UNIQUE(active_slot_id) rejects a second active booking for that slot, even if two requests arrive together. For a CANCELLED appointment, active_slot_id is NULL; H2 allows multiple NULLs in a unique column. The old row remains as history and the slot becomes bookable again. A future booking endpoint must turn a duplicate-key error into a clear conflict response.

Tutor schedule rule

Slots start on whole hours and last exactly one hour. UNIQUE(provider_id, starts_at) therefore prevents a tutor from offering overlapping slots, even for different services. Different tutors may offer sessions at the same time.

What runs in this milestone

Spring loads *schema.sql* and *seed.sql* at startup. The sample data has one student, two tutors, two services, and four future slots; slot 2 is already booked. **GET /** returns tutor and service DTOs. **GET /slots** returns slots 1, 3, and 4 as DTOs. The four required wireframes (Home, available slots, booking form, confirmation) are in *Wireframes.pdf*. Booking, login, notifications, and AI are planned features, not implemented Milestone 1 features.

Requirement source: SJSU CMPE 172, *Milestone 1 - Requirements, Design & Skeleton*, supplied two-page assignment PDF, pp. 1-2.